# Java Strings

Working with Text as Objects

A **String** in Java is an object that represents a sequence of characters. Unlike primitive types, Strings come with built-in methods for manipulation.

```
String greeting = "Hello World";
```

## 1. The length() Method

Find out exactly how many characters (including spaces) are in your text.

```
String txt = "ABCDE";
System.out.println(txt.length()); // Output: 5
```

## 2. Transforming Text

Quickly change the casing of your strings for comparisons or display.

> `toUpperCase()` → "hello" becomes "HELLO"

> `toLowerCase()` → "HELLO" becomes "hello"

## 3. Finding Positions (Index)

⚠️ **Crucial Rule:** Java starts counting at **0**.
In "Hello", 'H' is at index 0, 'e' is at 1, etc.

Use `indexOf()` to find a specific word or character:

```
String txt = "Locate the word";
System.out.println(txt.indexOf("the")); // Output: 7
```

Use `charAt()` to pick a character at a specific spot:

```
String myStr = "Java";
System.out.println(myStr.charAt(0)); // Output: J
```

### Visual Summary

- **Storage:** Text must be in **"Double Quotes"**.
- **Nature:** In Java, String is a **class**, not a primitive.
- **Counting:** Always start from **zero**.
- **Modification:** Strings are immutable (the original text doesn't change; the methods return a NEW string).